

MAT205: Abstract Algebra II

Algebraic Closure via Zorn's Lemma

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Part I

Statement and Construction

Motivation: From One Root to Every Root

L12 Part IV: every nonconstant $f \in F[x]$ has *some* root in *some* extension E/F .

But this produces one root at a time; the extension depends on the polynomial.

We want **all** roots of every polynomial in a single ambient field – for splitting fields, Galois groups, factorization.

Pushed to the limit: a field $\overline{F}$ in which every nonconstant $f \in F[x]$ has a root, with no algebraic excess.

Goal of this lecture. Such an $\overline{F}$ exists and is essentially unique. Existence uses Zorn's lemma.

Definition: Algebraically Closed Field

Definition. A field K is **algebraically closed** if every nonconstant polynomial $f \in K[x]$ has a root in K .

Equivalent characterization. Every nonconstant $f \in K[x]$ factors completely into linear factors:

$$f = c \prod_{i=1}^{\deg f} (x - \alpha_i), \quad c \in K^\times, \alpha_i \in K.$$

Example. $\mathbb{C}$ is algebraically closed (Fundamental Theorem of Algebra; classical proof outside this course).

Non-examples. $\mathbb{Q}, \mathbb{R}, \mathbb{F}_p$ – each has irreducible polynomials of degree ≥ 2 .

Definition: Algebraic Closure

Definition. An **algebraic closure** of a field F is a field $\overline{F}$ such that:

1. $F \subseteq \overline{F}$ (extension);
2. $\overline{F}/F$ is an **algebraic extension** (L12 Part VI: Definition of Algebraic Extension);
3. $\overline{F}$ is algebraically closed.

Reading. $\overline{F}$ contains a root of every nonconstant $f \in F[x]$, and nothing transcendental is added beyond what algebraicity forces.

Concrete example. $\overline{\mathbb{Q}} := \{\alpha \in \mathbb{C} : \alpha \text{ algebraic over } \mathbb{Q}\}$ is an algebraic closure of $\mathbb{Q}$. Algebraicity from L12 (Algebraic Elements Form a Subfield). Algebraic closedness: $\mathbb{C}$ is algebraically closed plus transitivity (L12: Transitivity of Algebraicity).

Theorem: Steinitz (Existence and Uniqueness)

Theorem (Steinitz, 1910). Every field F has an algebraic closure $\overline{F}$.

Furthermore, any two algebraic closures of F are isomorphic as extensions of F .

Both existence and uniqueness use **Zorn's lemma** (equivalent to the Axiom of Choice).

Historical note. Steinitz's 1910 paper systematized field theory using transfinite methods – one of the earliest serious uses of choice principles in algebra.

Proof Skeleton: The Zorn Argument

Step 1 (Bound the ambient set). All algebraic extensions of F have cardinality $\leq \max(|F|, \aleph_0)$. Fix an ambient set Ω of this cardinality with $F \subseteq \Omega$.

Step 2 (Define the poset). Let

$$\mathcal{P} = \{ (K, +_K, \cdot_K) : F \subseteq K \subseteq \Omega, K \text{ a field, } K/F \text{ algebraic} \},$$

partially ordered by inclusion.

Step 3 (Apply Zorn's lemma). Every chain has an upper bound (the union – a field, algebraic over F). Zorn yields a maximal element $\overline{F} \in \mathcal{P}$.

Step 4 (Maximal $\Rightarrow$ algebraically closed). If $f \in \overline{F}[x]$ has no root in $\overline{F}$, build $\overline{F}(\alpha) \supsetneq \overline{F}$ with $f(\alpha) = 0$ (L12: Theorem on Existence of a Root). Then α is algebraic over F by transitivity (L12: Transitivity of Algebraicity). Embed into Ω – contradicts maximality.

Conclusion. $\overline{F}$ is algebraically closed and algebraic over F – an algebraic closure of F .

Universal Property of Algebraic Closure

Theorem. Let $\overline{F}/F$ be an algebraic closure. For any algebraic extension E/F , there exists an F -embedding

$$\sigma : E \hookrightarrow \overline{F}, \quad \sigma|_F = \text{id}_F.$$

Slogan. $\overline{F}$ is "big enough" to contain a copy of every algebraic extension of F .

Non-uniqueness. For $E = F(\alpha)$ with α a root of irreducible $p \in F[x]$, the F -embeddings $E \hookrightarrow \overline{F}$ are in bijection with the roots of p in $\overline{F}$.

Corollary (Uniqueness of $\overline{F}$). Any two algebraic closures of F are F -isomorphic. Embed one closure into the other; by maximality the image is everything.

Proof sketch. Zorn on pairs $(K, \sigma_K : K \hookrightarrow \overline{F})$ with $F \subseteq K \subseteq E$, ordered by extension. At a maximal pair, $K = E$: otherwise pick $\beta \in E \setminus K$ and send it to any root of m_β in $\overline{F}$.